PROJECT PHASE 3: BASELINE MODEL

Deadline: Week 8

Weight: 15% of the project grade

Penalty: Each delay is penalized with 3 points

Maximum allowed delays: 2

🎯 Objective

The goal of this phase is to **build the first working machine learning model (baseline)** /or discovery pattern model using the processed data from Phase 2.

Students must:

- Implement a **complete training pipeline**
- Train a **first model (baseline)**
- Perform an **initial evaluation**
- Understand how data representation impacts model performance

This phase represents the **transition from data → model** and establishes a reference point for further improvements.

💻 Coding (Baseline model)

1. Data Preparation for Modelling
2. Data Representation
    a. You must clearly define how your data is represented for the model:
3. Baseline Model
4. The whole pipeline (training, validation, testing)
5. Evaluation
    a. Evaluate the model using appropriate metrics depending on the problem:
    b. Must include:
        i. Interpretation of results
        ii. What the model does well / poorly
6. **Error Analysis (VERY IMPORTANT)**
    You must analyze:
    - Where the model fails
    - Possible causes
    - Examples: Misclassified classes, Outliers affecting regression, Bias in predictions

This is critical for Phase 4 where you will perform improvements

Report

Please continue to work on the report template provided:
https://github.com/SergiuLimboi/Intelligent-techniques-for-processing-structured-and-large-data/tree/main/Report%20template

IMPORTANT:

- **You do NOT create separate reports for each phase**
- This phase, and all phases, must be **integrated into the SAME report (conclusions, introduction and abstract are written at the end of the project)**

The report must include:

---

1. Baseline Model

- Description of the chosen algorithm
- Why it was selected
- Assumptions of the model

2. Data Representation

- How the data is transformed for modeling
- Feature vector description
- Justification of representation

3. Model Implementation

- Description of training pipeline
- Key steps (preprocessing → training → prediction)
- Hyperparameters used

4. Evaluation

- Metrics used (with justification)
- Results (tables / plots)

Must include:

- Interpretation (not just numbers)
- Comparison (if multiple runs/settings)

5. Error Analysis

- Where the model fails
- Observations on errors
- Potential improvements

📦 Deliverables (Mandatory)

Same as previous phases:

Report:

- **LaTeX source**
- **PDF version**

Code:

- **Jupyter Notebook / Python scripts**
- **Must be runnable**
- **Must include explanations**

Repository:

- **GitHub Classroom (same repo)**

Submission rule:

Upload: Before laboratory in Week 8

📊 Grading

- Report: **50%**
- Implementation (code): **50%**

Final grade = weighted average

Grades will be provided after evaluation by course & lab instructors

❗ Important

- Work will NOT be evaluated if not committed to GitHub Classroom
- Ensure all required files are present before deadline

Grading Rubric — Code (Baseline model)

Section	Points	Full points	Partial points	No points
1 . Data representation	1.5	Clear representation + justification	Representation unclear	Missing
2.Model implementation	2	Correct algorithm and explanation of used parameters	poorly structured	Missing
3 . Learning pipeline/discovery pattern	2	Clear, reproducible pipeline .	Pipeline unclear	Missing
4 . Evaluation	1.5	Correct metrics + interpretation	Metrics without interpretation	Missing
5 . Error analysis	2	Deep analysis of errors	Superficial	Missing
6 . Code clarity	1	Clean, documented	Superficial	Poor
Total	**10 points**			

Grading Rubric — Report Phase 3

Section	Points	Full points	Partial points	No points
1 . Baseline model	2	Well explained + justified	Basic description	Missing
2 . Data representation	2	Clear + justified	Unclear	Missing
3 . Model pipeline	2	Structured explanation (diagrams, details) .	Partial	Missing
4 . Evaluation & error analysis	3	Metrics + interpretation + Insightful analysis	Superficial	Missing
5 . Clarity & structure	1	Well written, structured	Acceptable	Poor
Total	**10 points**			